1. Coordinate Geometry

1.1. Graphs

Gradient of a Straight Line

- Measure of the steepness of a straight line
- $Gradient = \frac{y_2 - y_1}{x_2 - x_1}$

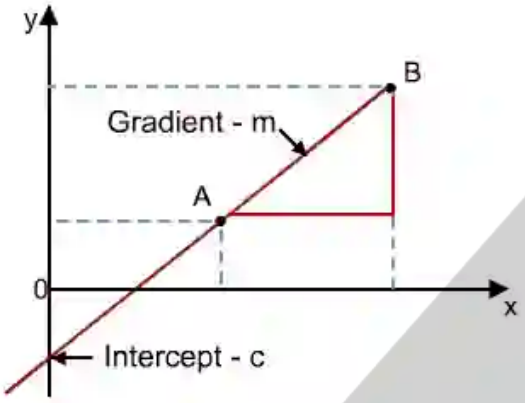
Equation of a Line

$$Ax + By = C \quad \text{(General Equation of Line)}$$

$$\frac{x}{a} + \frac{y}{b} = 1 \quad \text{(Intercept Form)}$$

$$y = mx + c \quad \text{(Slope–Intercept Form)}$$

- m = gradient
- c = y-intercept (the point the graph meets the y-axis)



Midpoint of Graph

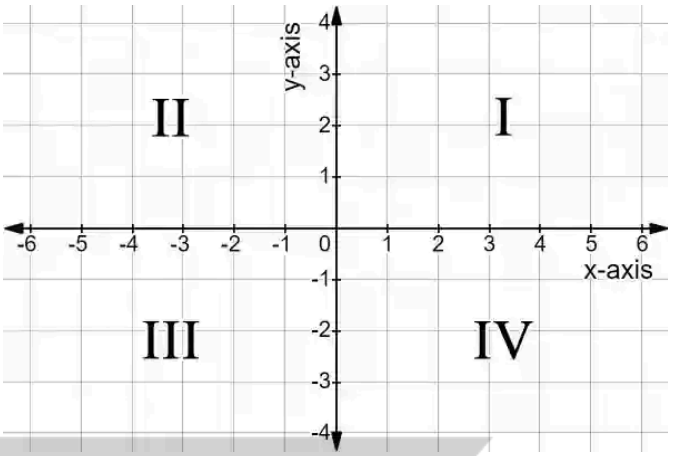
$$Midpoint = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$

Length between Two Points

$$Distance = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

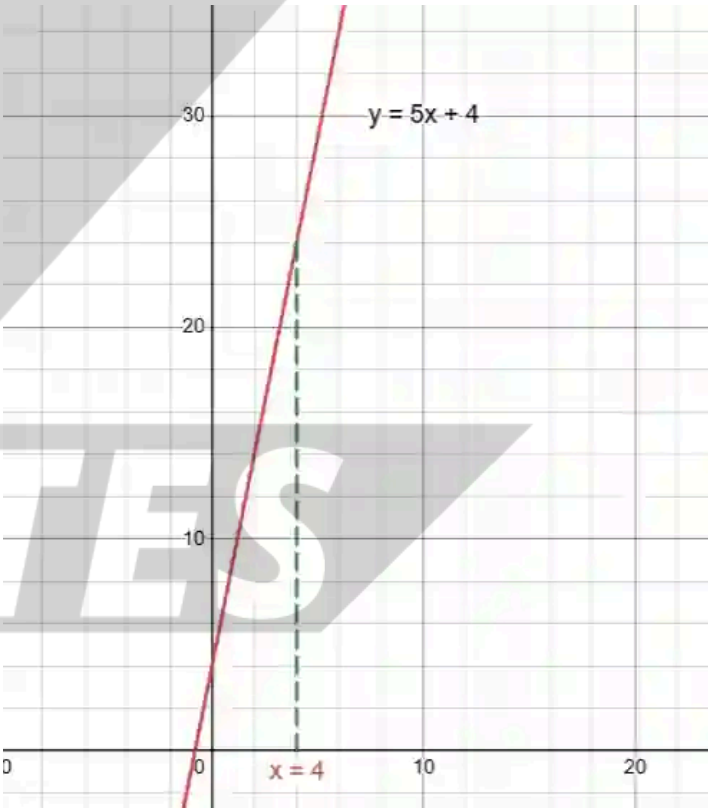
Cartesian Coordinate System

- They are written in the (x,y) for



- **Quadrant 1:** (x,y)
- **Quadrant 2:** (-x,y)
- **Quadrant 3:** (-x,-y)
- **Quadrant 4:** (x,-y)

1.2. Sketching Graphs



- **Step 1:** Plot the y-intercept (c) on the graph
- **Step 2:** Substitute x values (whole number values) into the equation and solve for y
- **Step 3:** The substituted x value is the x coordinate, and the solved y value is the y coordinate
- **Step 4:** Repeat for 4 more values and draw a straight line connecting these points

How to Derive Equation of Line from Graph

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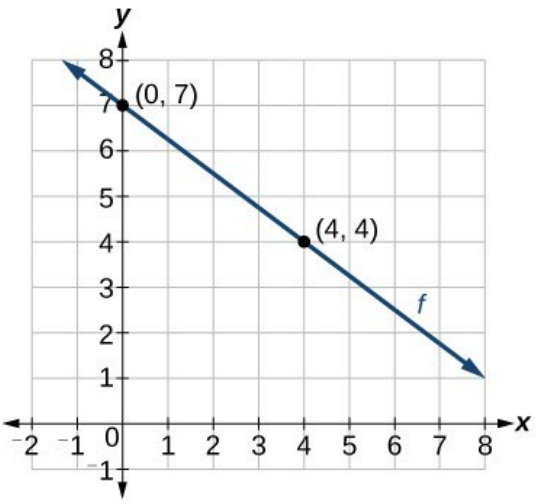


Find the gradient of $(0,7)$ and $(4,4)$:

$m = \frac{y_2-y_1}{x_2-x_1}$

$m = \frac{4-7}{4-0}$

$m = \frac{-3}{4}$

Equation of the line: $y = mx + c$

$y = -\frac{3}{4}x + c$

Substitute $(0,7)$:

$7 = -\frac{3}{4}(0) + c$

$7 = c$

Therefore, $y = -\frac{3}{4}x + 7$

1.3. Parallel Lines and Perpendicular Bisectors

Parallel Lines

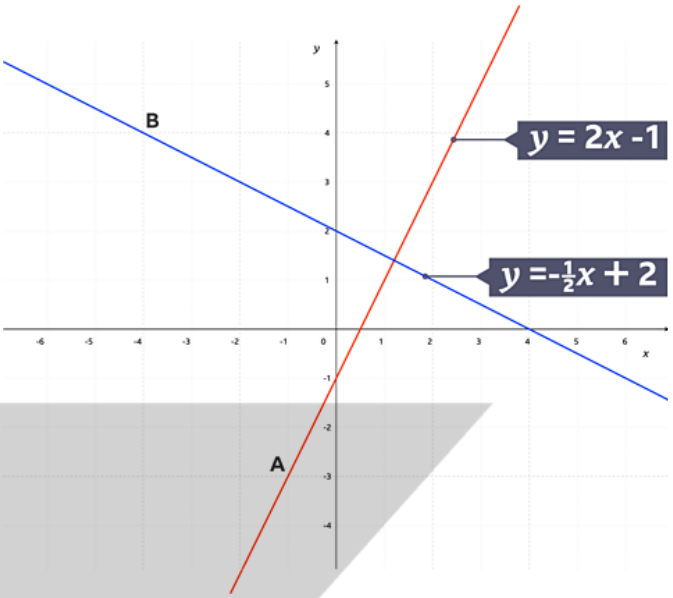
- Two lines are parallel if they share the same gradient but different y-intercepts
- For eg, $y = 3x + 5$ and $y = 3x + 12$
- Since the gradient of both lines is the same, and they have different y-intercepts, they are parallel to each other

Perpendicular Lines

- Two Lines are perpendicular if they meet at 90°
- If Line A has a gradient m, then the gradient of Line B is -1/m
- For eg, $y = 2x + 4$ and $y = \frac{-1}{2}x + 5$
- Two lines are perpendicular if the gradient of one line is the negative reciprocal of the other

Perpendicular Bisectors

- Perpendicular Bisector is a line that intersects another line at its midpoint, and lies at an angle of 90°



- Example:

  Point A has coordinates $(7,-17)$

  Point B has coordinates $(19,-11)$

  We want the equation of the perpendicular bisector of AB .

  Step 1: Midpoint

  $M = \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$

  $M = \left(\frac{7+19}{2}, \frac{-17+(-11)}{2}\right)$

  $M = \left(\frac{26}{2}, \frac{-28}{2}\right)$

  $M = (13,-14)$

  Step 2: Gradient of AB

  $m_{AB} = \frac{y_2-y_1}{x_2-x_1}$

  $m_{AB} = \frac{-11-(-17)}{19-7}$

  $m_{AB} = \frac{6}{12} = \frac{1}{2}$

  Step 3: Gradient of perpendicular bisector

  $m_{\perp} = -\frac{1}{m_{AB}} = -2$

  Step 4: Equation of perpendicular bisector

  $y - (-14) = -2(x - 13)$

  $y + 14 = -2x + 26$

  $y = -2x + 12$

  Therefore, the equation of the perpendicular bisector is:

  $\boxed{y = -2x + 12}$

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